

Quantum Mechanics-II

PHYS:5742

The University of Iowa
The College of Liberal Arts and Sciences
Spring, 2026

Course meeting time and place

9:30 am -- 10:20 am Mondays, Wednesdays, Fridays, VAN 161

Department of Physics & Astronomy

<https://physics.uiowa.edu/>

Instructor: Prof. Matheus Hostert

Phone: 319-335-2943

Email: matheus-hostert@uiowa.edu

Office location: VAN 502

Student drop-in hours

10:00 am -- 11:00 am Tuesday

3:00 pm -- 4:00 pm Wednesday

2:30 pm -- 3:30 pm Thursday

Or by appointment

Departmental Executive Officer (DEO): Gregory G. Howes (gregory-howes@uiowa.edu)

Student Complaints

Students with a complaint about a grade or a related matter should first discuss the situation with the instructor and/or the course supervisor (if applicable), and finally with the DEO (Chair) of the department, school or program offering the course. Sometimes students will be referred to the department or program's Director of Undergraduate Studies (DUS) or Director of Graduate Studies (DGS).

Undergraduate students should contact [CLAS Undergraduate Programs](#) for support when the matter is not resolved at the previous level. Graduate students should contact the [CLAS Graduate Affairs Manager](#) when additional support is needed.

Course's College (Administrative Home)

For undergraduate courses

The College of Liberal Arts and Sciences (CLAS) is the home of this course, and CLAS governs the add and drop deadlines, academic misconduct policies, and other undergraduate policies and procedures. Other UI colleges may have different policies.

For graduate courses

The College of Liberal Arts and Sciences (CLAS) is the home of this course, and CLAS governs the policies and procedures for its courses. Graduate students, however, must adhere to the [academic deadlines set by the Graduate College](#).

Drop Deadline for this Course

You may drop an individual course before the drop deadline; after this deadline you will need collegiate approval. You can look up the drop deadline for this course [here](#). When you drop a course, a “W” will appear on your transcript. The mark of “W” is a neutral mark that does not affect your GPA. To discuss how dropping (or staying in) a course might affect your academic goals, please contact your Academic Advisor. Directions for adding or dropping a course and other registration changes can be found on the [Registrar’s website](#). Undergraduate students can find policies on dropping CLAS courses [here](#). Graduate students should adhere to the [academic deadlines](#) and policies set by the Graduate College.

UI Email

Students are responsible for all official correspondences sent to their UI email address (uiowa.edu) and must use this address for any communication with instructors or staff in the UI community. For the privacy and the protection of student records, UI faculty and staff can only correspond with UI email addresses.

Description of Course

This course is a continuation of PHYS:5741. Topics to be covered include Chapters 5 – 8 of Sakurai & Napolitano. Please see the calendar below for more information.

Learning Outcomes

Students will be able to solve perturbative problems in Quantum Mechanics with time-independent and dependent Hamiltonians, calculate rates and scattering cross sections, analyze multi-particle systems, and will be familiar with the solutions to the Klein-Gordon and Dirac equations.

Textbooks/Materials

The required textbook(s) for this course are:

Modern Quantum Mechanics (edition: 3rd) Required

ISBN: 9781108473224

Author: Sakurai

Publisher: CAMBRIDGE UNIV PRESS. ©2020

Approximate Price: \$54.75

Course ICON site

To access the course site, log into [Iowa Courses Online \(ICON\)](#) using your Hawk ID and password.

Grading System

Possible final grades are A+ to F.

B represents a minimal satisfactory learning, **A** an outstanding performance, and **C** and below a substandard performance indicating that the completion of a PhD degree may be difficult. Plus/minus grades may be given. No curve involved. **A+** grades are given only in extraordinary situations.

Homework assignments, solutions, and notes will be posted on ICON.

Course Grades

Homework 20%, Midterm-1 25%, Midterm-2 25%, Final Exam 30%.

Homework assignments will be set every week with exceptions to be announced on ICON. Each homework contributes equally to the final homework grade. Homework will be self-graded, and the grading will be reviewed by the instructor.

Homework assignments will be set on Fridays and will be due on the next **Friday by 5 pm**. Please check announcements on ICON for more details.

Late submissions will be penalized by 10 percentage points within the first 24 hours, and 20 percentage points within the second 24 hours. No assignments will be accepted after 48 hours, including weekends.

Attendance and Accommodations

Regular attendance is encouraged but not enforced. While attendance is not required, you will be responsible for all the material covered in the lectures.

Questions during class are encouraged, no matter how trivial, however, the instructor may recommend some discussions to be continued during office hours. Please bring your own writing material to solve exercises during class.

Absences from Class

University regulations require that students be allowed to make up examinations which have been missed due to illness, religious holy days, military service obligations, including

service-related medical appointments, jury duty, or other unavoidable circumstances or other university-sponsored activities. Students should work with their instructor regarding making up other missed work, such as assignments, quizzes, and classroom attendance.

Absences for Religious Holy Days

The university is prepared to make reasonable accommodations for students whose religious holy days coincide with their classroom assignments, test schedules, and classroom attendance expectations. Students must notify their instructors in writing of any such religious holy day conflicts or absences within the first few days of the semester or session, and no later than the third week of the semester. If the conflict or absence will occur within the first three weeks of the semester, the student should notify the instructor as soon as possible. See [Policy Manual 8.2 Absences for Religious Holy Days](#) for additional information.

Absences for Military Service Obligations

Students absent from class or class-related requirements due to U.S. veteran or U.S. military service obligations (including military service–related medical appointments, military orders, and National Guard Service obligations) shall be excused without any grading adjustment or other penalty. Instructors shall make reasonable accommodations to allow students to make up, without penalty, tests and assignments they missed because of veteran or military service obligations. Reasonable accommodations may include making up missed work following the service obligation; completing work in advance; completing an equivalent assignment; or waiver of the assignment without penalty. In all instances, students bear the responsibility to communicate with their instructors about such veteran or military service obligations, to meet course expectations and requirements.

Accommodations for Students with Disabilities

The University is committed to providing an educational experience that is accessible to all students. If a student has a diagnosed disability or other disabling condition that may impact the student's ability to complete the course requirements as stated in the syllabus, the student may seek accommodations through [Student Disability Services](#) (SDS). SDS is responsible for making [Letters of Accommodation \(LOA\)](#) available to the student. **The student must provide an LOA to the instructor as early in the semester as possible, but requests not made at least two weeks prior to the scheduled activity for which an accommodation is sought may not be accommodated.** The LOA will specify what reasonable course accommodations the student is eligible for and those the instructor should provide. Additional information can be found on the [SDS website](#).

Students should feel free to get in touch via email or during office hours to discuss any accommodations necessary, as outlined above.

Other Expectations of Student Performance

Free Speech and Expression

The University of Iowa supports and upholds the First Amendment protection of freedom of speech and the principles of academic and artistic freedom. We are committed to open inquiry, vigorous debate, and creative expression inside and outside of the classroom. Visit the [Free Speech at Iowa website](#) for more information on the university's policies on free speech and academic freedom.

Non-discrimination Statement

The University of Iowa prohibits discrimination in employment, educational programs, and activities on the basis of race, creed, color, religion, national origin, age, sex, pregnancy (including childbirth and related conditions), disability, genetic information, status as a U.S. veteran, service in the U.S. military, sexual orientation, or associational preferences. The university also affirms its commitment to providing equal opportunities and equal access to university facilities. For additional information on nondiscrimination policies, contact the Senior Director, Office of Civil Rights Compliance, the University of Iowa, 202 Jessup Hall, Iowa City, IA 52242-1316, 319-335-0705, ui-ocrc@uiowa.edu. Although not required, students have the option to share their pronouns and chosen/preferred names in class and through MyUI. Instructors and advisors can find information about a student's chosen/preferred name in MyUI.

Classroom Expectations

Students are expected to comply with University policies regarding appropriate classroom behavior as outlined in the [Code of Student Life](#). While students have the right to express themselves and participate freely in class, it is expected that students will behave with the same level of courtesy and respect in the virtual class setting (whether asynchronous or synchronous) as they would in an in-person classroom. Failure to follow behavior expectations as outlined in the [Code of Student Life](#) may be addressed by the instructor and may also result in discipline under the [Code of Student Life](#) policies governing E.5 Disruptive Behavior or E.6 Failure to Comply with University Directive.

Class Recordings

The unauthorized video or audio recording of academic activities (e.g., lectures, course discussions, office hours, etc.) by a student is prohibited. Students with a reasonable accommodation for recording approved by Student Disability Services should notify each instructor and provide the Letter of Accommodation prior to using the accommodation. A student may record classroom activities with prior written permission from the instructor and notice to other students in the class that audio or video recording may occur. Any and all classroom recording must be for personal academic use only. The distribution, sharing, sale, or posting of recordings on the internet (including social media), in whole or in part, is prohibited and doing so may be a violation of the Code of Student Life and/or state or federal privacy, copyright, or other laws.

Academic Honesty and Misconduct

All students in CLAS courses are expected to abide by the [college's standards of academic honesty](#). Undergraduate academic misconduct must be reported by instructors to CLAS according to [these procedures](#). Graduate academic misconduct must be reported to the Graduate College according to Section F of the [Graduate College Manual](#).

Collaborative work is encouraged, but plagiarism is not. Students are responsible for ensuring their own academic honesty and encouraged to ask their instructor for clarification if needed.

Artificial Intelligence (AI) Policy

Usage of AI tools such as ChatGPT, Copilot, or Claude, is allowed and encouraged for coding, idea generation, or brainstorming, but it should not be used to solve your homework and should not be used blindly as hallucinations are common and can look convincing. Opt for books and resources provided by the instructor to learn new material. If using AI tools, note that “thinking” or “reasoning” models are not necessarily less likely to hallucinate. See for example [this OpenAI article](#) and [this Claude article](#). AI tools should not be used for generating solutions or code you do not understand.

If AI is used in a homework or assignment, include a brief “AI usage” note explaining what you used it for and why you believe it is correct.

Date and Time of the Final Exam

1st Midterm: Friday, March 6th, 2026

2nd Midterm: Friday, April 24th, 2026

Final exam: Monday, May 11th, 2026

The [final examination date and time](#) will be announced by the Registrar generally by the fifth week of classes, and it will be announced on the course ICON site once it is known. **Do not plan your end of the semester travel plans until the final exam schedule is made public. It is your responsibility to know the date, time, and place of the final exam.** The [Registrar's Office website](#) details which types of schedule conflicts qualify for make-up exams and the processes students should follow to request make-up exams. Not all conflicts qualify for make-up exam times, and requests must be made by the deadlines listed.

Calendar of Course Assignments and Exams

Please note this calendar is **a tentative plan and subject to change**. Ranges are inclusive.

Topics Covered	Dates
Module 1: Review and WKB (2 weeks)	
Solutions to Schrödinger's equation, Dirac notation and wavefunctions, time-evolution, and the WKB approximation.	Weeks 1 – 2 (Jan 19 th – Jan 30 th) <i>MLK day (No classes on Jan 19th)</i>
Module 2: Perturbation theory (4 weeks)	
Time-independent and time-dependent perturbation theory. Examples and degenerate cases. Sudden and adiabatic limits of time-dependent problems. Dyson series and applications.	Weeks 3 – 6 (Feb 2 nd – Feb 27 th)
Module 3: Scattering theory (4 weeks)	
Rates and cross sections, the Born approximation, partial waves, and resonances. Spring Break (no classes during Week 9: Mar 16 -- 20)	Weeks 7 – 10 (Mar 2 nd – Mar 27 th) Midterm 1: March 6th
Module 4: Identical Particles (2 weeks)	
Fermions and bosons, multiparticle systems, and a first look at second quantization.	Weeks 11 – 12 (Mar 30 th – Apr 10 th)
Module 5: Relativistic Quantum Mechanics (3 weeks)	
Relativistic systems and the Klein-Gordon and Dirac equations. Spinors, Dirac algebra, and particle-antiparticle interpretation.	Weeks 13 – 15 (Apr 13 th – May 1 st) Midterm 1: April 24th
Overflow & Review (1 week)	(May 4 th – May 8 th)
Final Exam	Monday, May 11th, 2026

Student Support Resources and Related Policies

Academic Support for this Course

I am available for help during drop-in hours, via email, or by appointment. Please feel free to reach out. You may also access academic support via Tutor Iowa at <https://tutor.uiowa.edu/> and writing support via The Writing Center at <https://writingcenter.uiowa.edu/>.

Mental Health

Students are encouraged to be mindful of their mental health and seek help as a preventive measure or if feeling overwhelmed and/or struggling to meet course expectations. Students are encouraged to talk to their instructor for assistance with specific class-related concerns. For additional support and counseling, students are encouraged to contact University Counseling Service (UCS). Information about UCS, including resources and how to schedule an appointment, can be found at counseling.uiowa.edu. Find out more about UI mental health services at: mentalhealth.uiowa.edu.

Basic Needs and Student Support

It can be difficult to maintain focus and be present if you are experiencing challenges with meeting basic needs or navigating personal crisis situations. The Office of the Dean of Students can help. Contact us for one-on-one support, identifying options, and to locate and access basic needs resources (such as food, rent, childcare, etc.).

Student Care and Assistance: 132 IMU, dos-assistance@uiowa.edu, 319-335-1162

Basic Needs info:

- [Food Pantry at Iowa](#)
- [Clothing Closet](#)
- [Basic Needs and Support Resources](#)

Sexual Harassment/Sexual Misconduct and Supportive Measures

The University of Iowa prohibits all forms of sexual harassment, sexual misconduct, and related retaliation. The [Policy on Sexual Harassment and Sexual Misconduct](#) governs actions by students, faculty, staff, and visitors. Incidents of sexual harassment or sexual misconduct can be reported to the [Office of Civil Rights Compliance](#) or to the [Department of Campus Safety](#). Students impacted by sexual harassment or sexual misconduct may be eligible for academic supportive measures and can learn more by [contacting the Office of Civil Rights Compliance](#). Information about confidential resources can be found on the [Office of Civil Rights Compliance website](#).

Conflict Resolution

The Office of the Ombudsperson is a confidential, impartial, informal, and independent resource for any member of the university community with a problem or concern. The Office of

the Ombudsperson offers a safe place to discuss conflicts or concerns. Students are encouraged to reach out for assistance. The office will brainstorm with students to help identify options, answer any questions, and provide referrals to other offices as appropriate. More information about the Office of the Ombudsperson, including how to schedule an appointment, can be found at ombudsperson.org.uiowa.edu.